



Lecture 5: Diffusion current, PN junction

ECE 340 – Electronic Circuits I Sec. 1

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(textbook sections 3.3-3.5)



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Previously..

Turning ON an electric field

Current flow type 1: Drift

Charges move due to Electric Field

Avg. velocity in E-field:

$$|\vec{v}_{n,p}| = \frac{q\vec{E}\tau_c}{m^*}$$

v-to-E proportionality constant is the **carrier mobility**: $|\vec{v}_{n,p}| = \mu_{n,p}\vec{E}$

$$\mu_{n,p} = \frac{q\tau_c}{m_{n,p}^*} \quad \ll \text{This is very important!}$$

For electrons: $v_{n,drift} = -\mu_n E$

For holes: $v_{p,drift} = \mu_p E$

The force on an individual charge

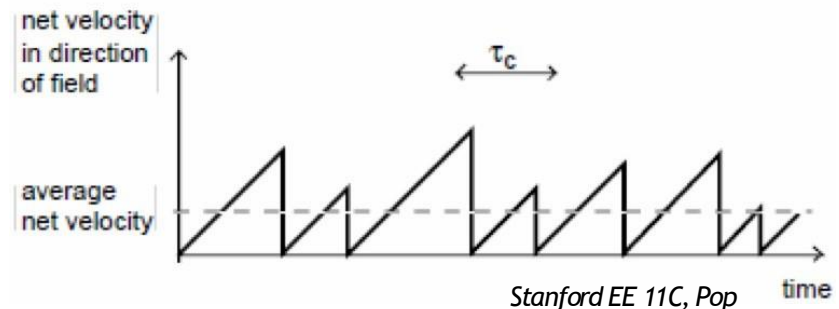
- $F = \pm qE$

- $F = m^*a \rightarrow a = \pm qE/m_{n,p}^*$

Between collisions, carriers accelerate along E field:

- $v_n(t) = a_n t = -qEt/m_n$ for electrons
- $v_p(t) = a_p t = +qEt/m_p$ for holes

but when they bump into the lattice, they slow down again!
(every τ_c collision time, velocity is random)



Mobility

$$\mu_{n,p} = \frac{q\tau_c}{m_{n,p}^*}$$

Mobility is a measure of how easily carriers **drift** in E-field

As m decreases, (particle is lighter), μ ...

As τ_c increases, (longer time b/w collisions), μ ...

	Si	Ge	GaAs	InAs	Graphene / CNTs
μ_n (cm ² /V·s)	1400	3900	8500	30000	40000
μ_p (cm ² /V·s)	470	1900	400	500	40000

electrons are more mobile!

Current density: $\vec{J}_n = -q\vec{n}\vec{v}_{n,drift} = qn\mu_n\vec{E} \left[\frac{A}{cm^2} \right]$

$$\vec{J}_p = qp\vec{v}_{p,drift} = qp\mu_p\vec{E} \left[\frac{A}{cm^2} \right]$$

$$\vec{J} = \vec{J}_n + \vec{J}_p = q(n\mu_n + p\mu_p)\vec{E}$$

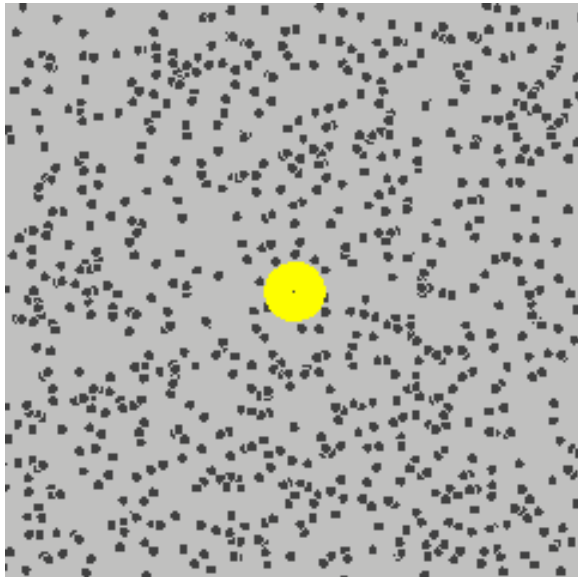
Conductivity, σ

Units are in cm²/ V·s :

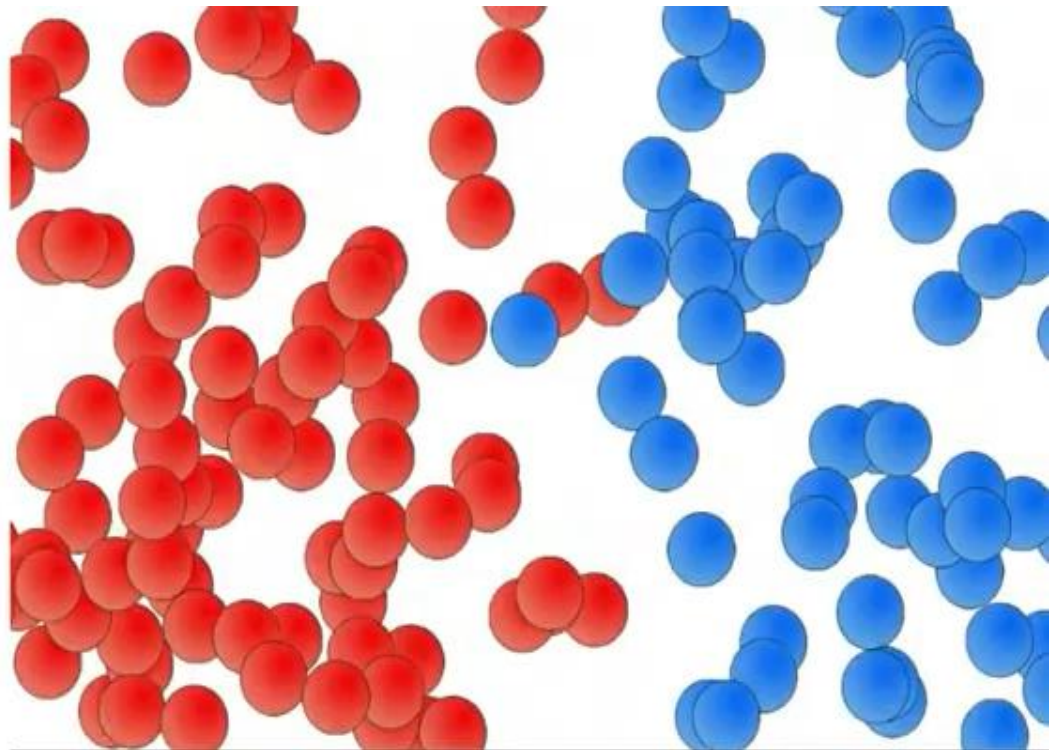
$$\frac{|\vec{v}_{n,p}|}{\vec{E}} = \mu_{n,p}$$

$$\frac{\left[\frac{cm}{s} \right]}{\left[\frac{V}{cm} \right]} = \left[\frac{cm^2}{V \cdot s} \right]$$

Current type 2: Diffusion current



Simulation of pollen (large particle) pushed around by atoms in a fluid

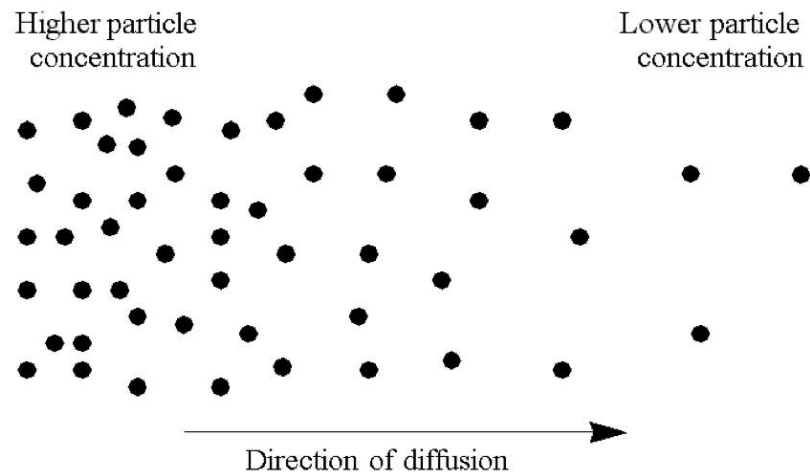


Current type 2: Diffusion current

- Remember Brownian motion of electrons & holes!
- When E-field = 0, but $T > 0$, thermal velocity $v_T = \underline{\sim 10^5 \text{ m/s}}$
- But net drift velocity $v_d = \underline{0}$
- So net current $J_d = \underline{0}$

Brownian motion:
"A random walk"

- What if there is
a *concentration or*
temperature (thermal
velocity) gradient?



Diffusion in semiconductors

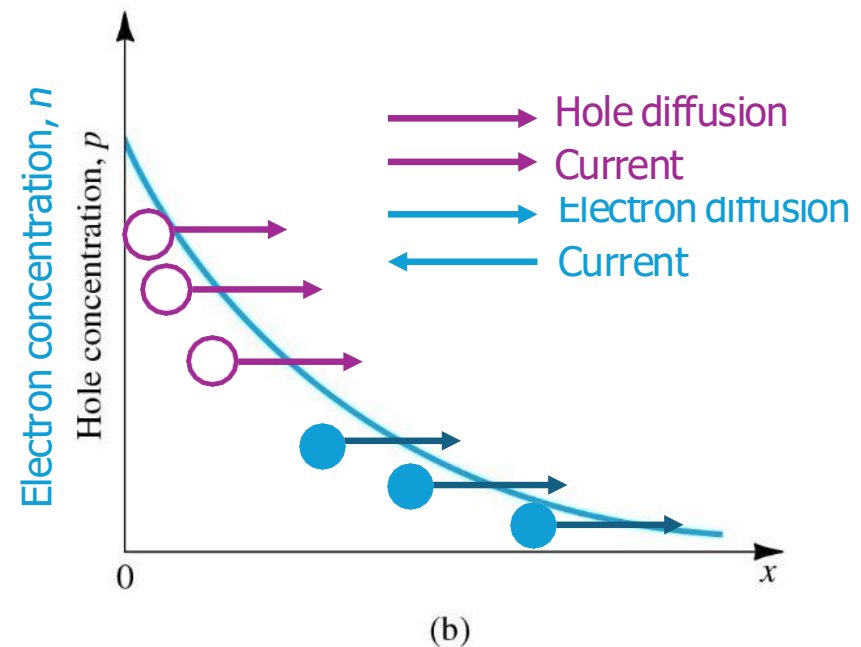
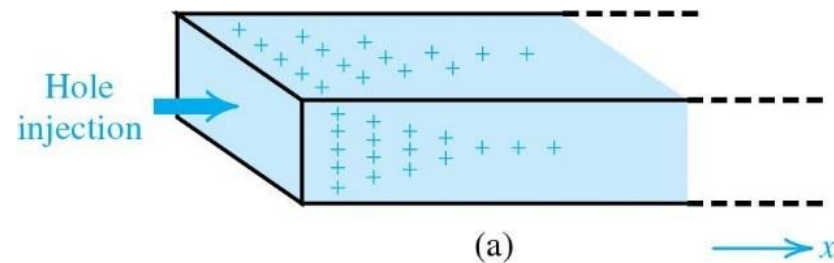
Starts with a difference in concentration of carriers across the material...

Current densities defined by:

$$J_{D,p} = -qD_p \frac{dp(x)}{dx} \left[\frac{A}{cm^2} \right]$$

$$J_{D,n} = qD_n \frac{dn(x)}{dx} \left[\frac{A}{cm^2} \right]$$

D_n, D_p are the diffusivities (diffusion constants) of electrons, holes. Typical values are $D_n = 35 \frac{cm^2}{s}$, $D_p = 12 \frac{cm^2}{s}$



Total currents in a semiconductor

- Now, we can FINALLY write down the TOTAL currents...

- For electrons: $J_N = J_{N,\text{drift}} + J_{N,\text{diff}} = qn\mu_n\mathcal{E} + qD_N \frac{dn}{dx}$

- For holes: $J_P = J_{P,\text{drift}} + J_{P,\text{diff}} = qp\mu_p\mathcal{E} - qD_P \frac{dp}{dx}$

- And TOTAL current:

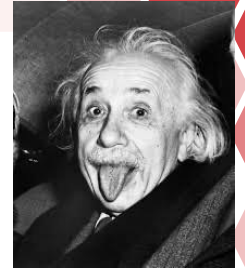
$$J_{\text{TOT}} = J_N + J_P$$

**minority carriers contribute little to drift current (usually, too few of them!), BUT if their gradient is high enough...

Einstein relationship

$$\frac{D}{\mu} = \frac{k_B T}{q} = V_T$$

<< This is very important!



$k_B = 8.617 \times 10^{-5} \text{ eV/K}$
or
 $k_B = 1.38 \times 10^{-23} \text{ J/K}$

it connects diffusivity with mobility,
which we already know how to look up.
(In other words, it describes how quickly
a particle will move in a medium due to a
concentration gradient.)

T = temperature [K]

V_T = Thermal voltage $\sim 25.9 \text{ mV}$ at $T = 300 \text{ K}$

***pn*-junction (no applied voltage)**

doped: p-type



in equilibrium

doped: n-type



"Quasi-neutral" = For the scales we're dealing with, the semiconductor is almost electrically neutral.

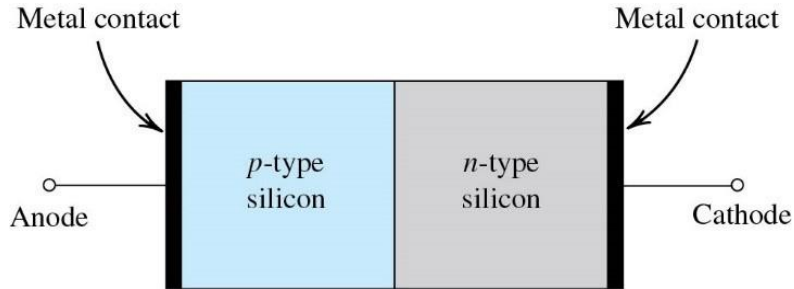
$p = N_A$ (acceptors are slightly negative)

Fixed atoms
Mobile charge carriers

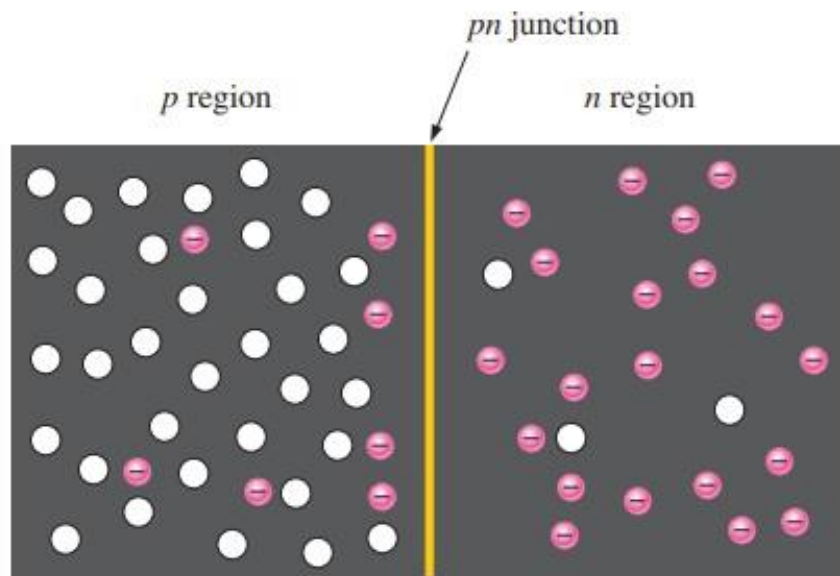
$n = N_D$ (donors are slightly positive)

Fixed atoms
Mobile charge carriers

Formation of *pn*-junction



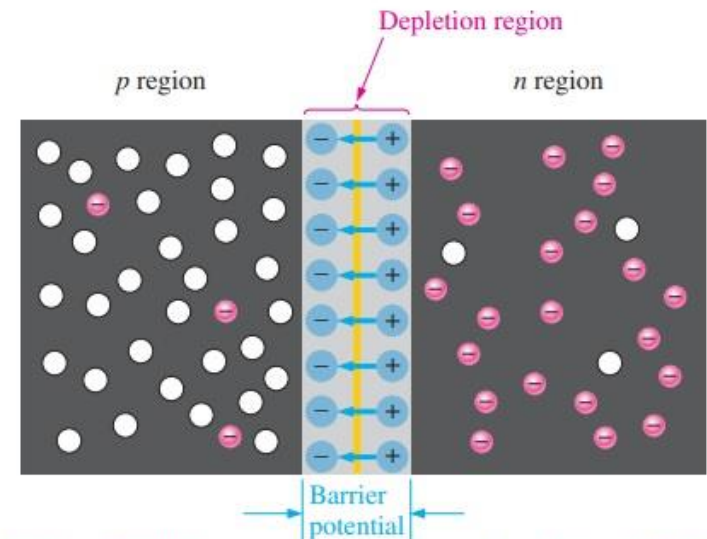
What is PN Junction



The basic silicon structure at the instant of junction formation showing only the majority and minority carriers. Free electrons in the n region near the pn junction begin to diffuse across the junction and fall into holes near the junction in the p region

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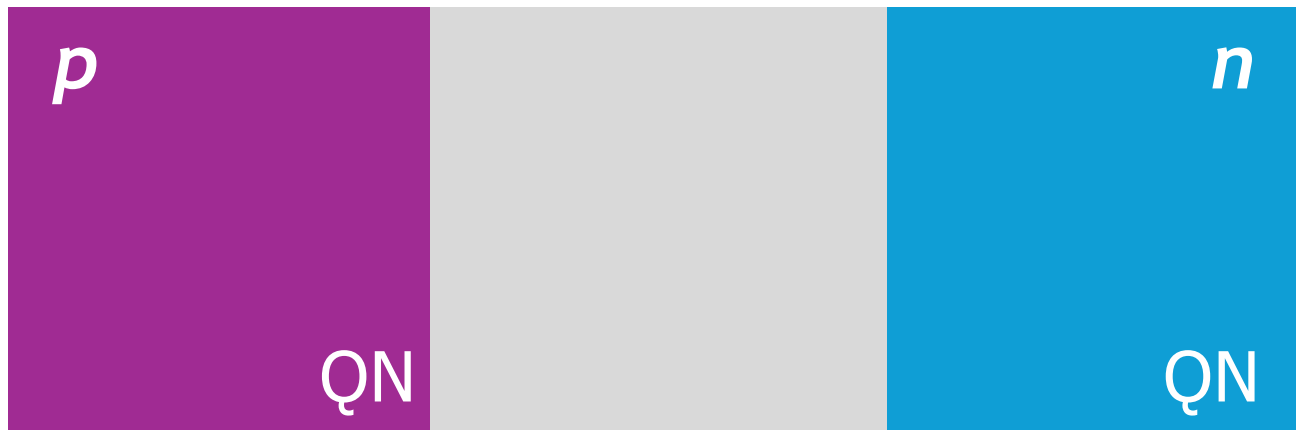
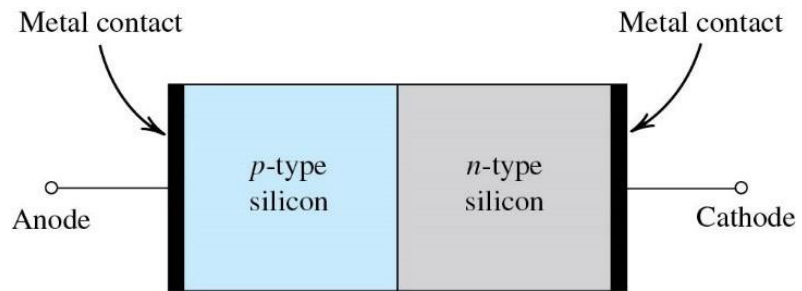
Formation of the Depletion Region



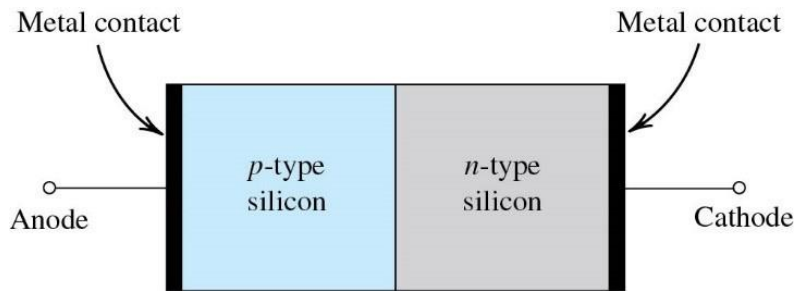
For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the n region and a negative charge is created in the p region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field

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***pn*-junction (no applied voltage)**

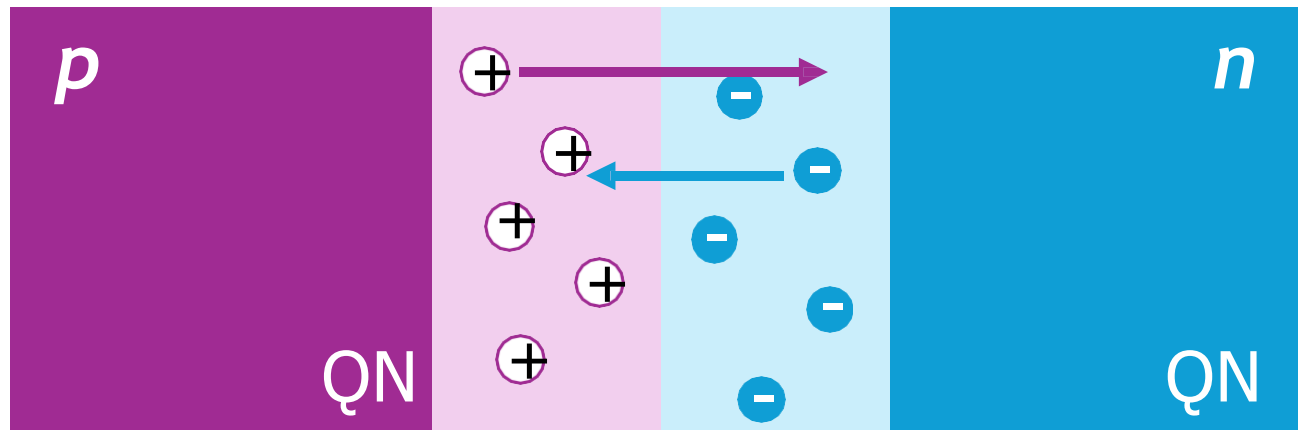


pn-junction (no applied voltage)



The holes recombine with some **majority electron carriers** on the **right**, causing a *depletion of free electrons* close to the junction

The electrons recombine with some **majority hole carriers** on the **left**, causing a *depletion of holes* close to the junction



I_D = diffusion current

pn-junction (no applied voltage)

Net charge density in depletion region is

$$q(N_D - N_A)$$

Only made up of fixed charge sites (N_A and N_D)

There are NO mobile charges in the depletion region

Electric field



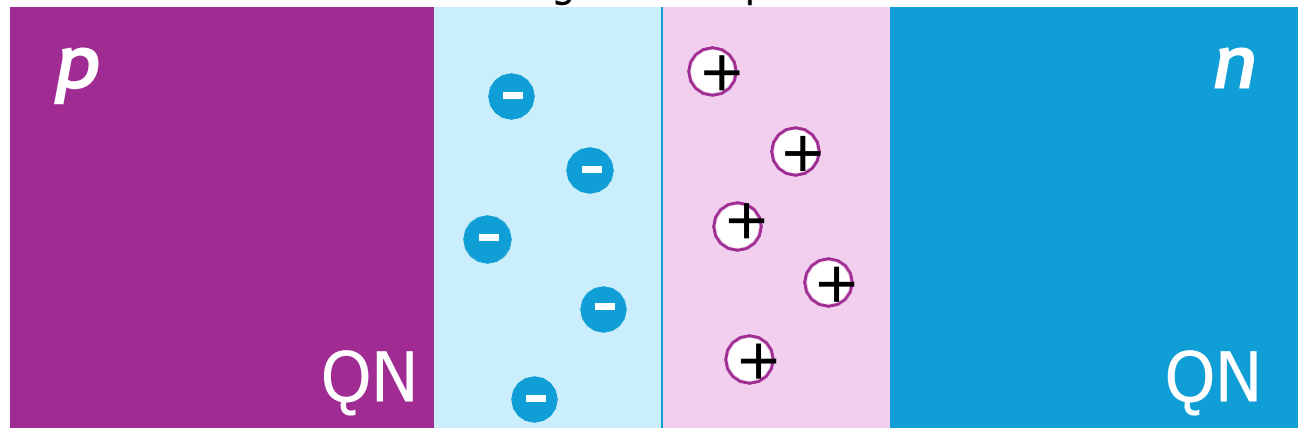
Depletion region,
"space-charge region"

net negative

net positive

The electric field opposes the diffusion of more holes into n region and more electrons into p region

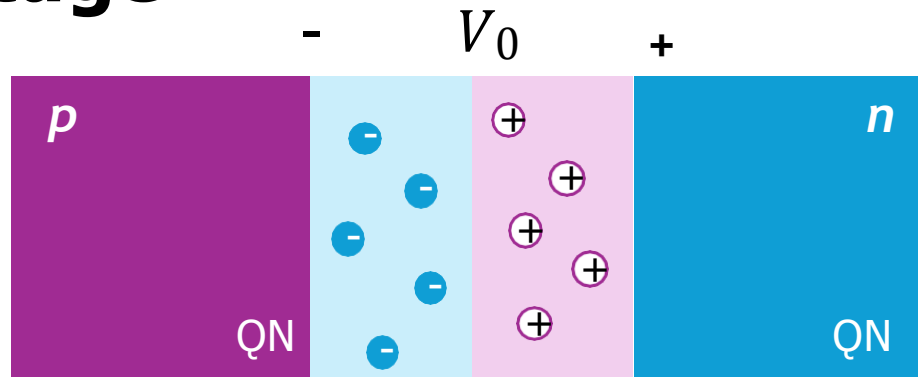
A voltage potential is established!!!



$I_D =$ diffusion current
(majority carrier diffusion)

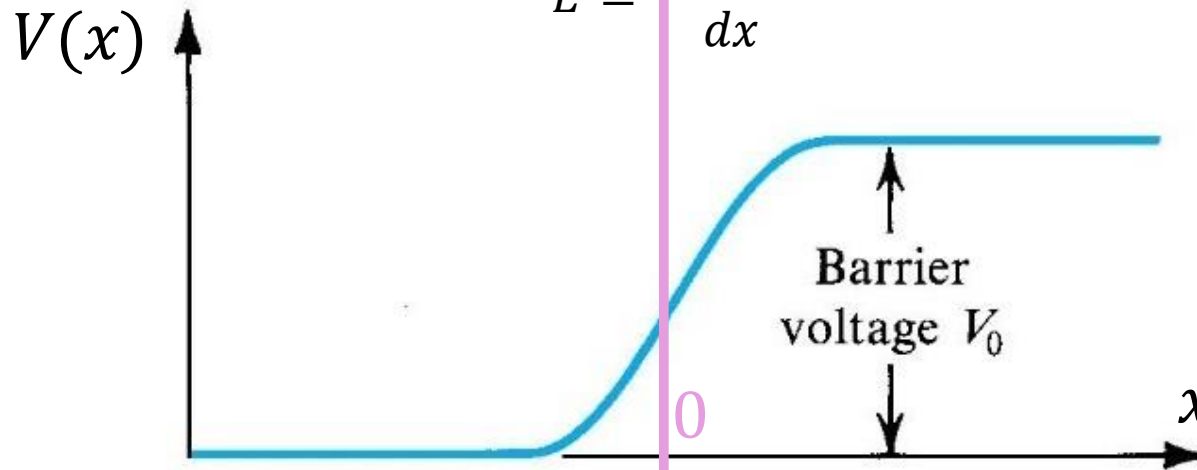
Built-in voltage

V_0 CANNOT be measured as an open-circuit voltage!
(opposing charge carriers at the contacts counteract this)



Electric field

$$E = -\frac{dV}{dx}$$



Voltage potential

With no voltage applied, barrier voltage is:

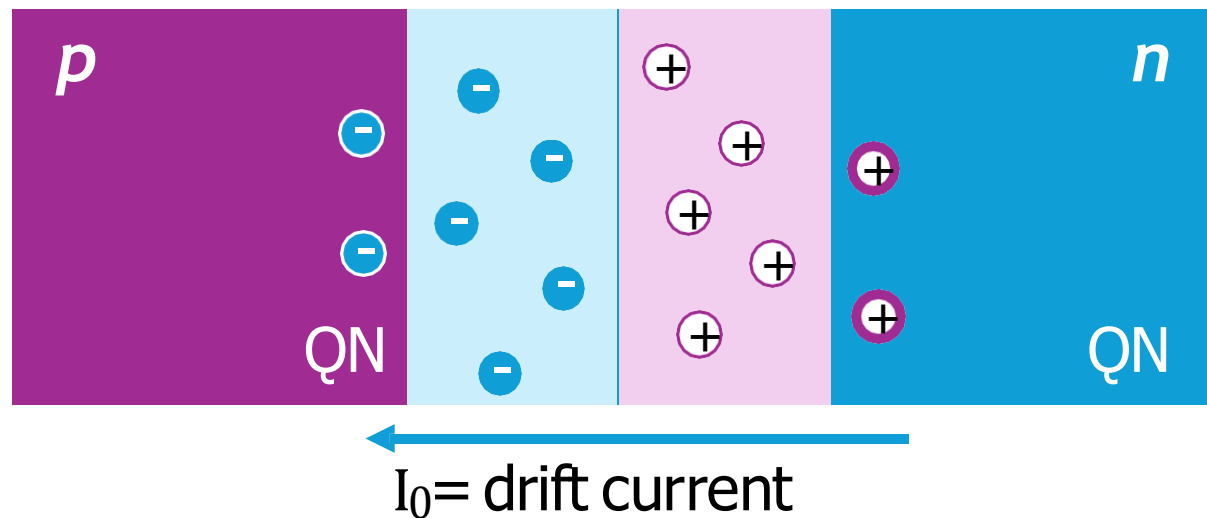
$$V_0 = V_T \ln \left(\frac{N_A N_D}{(n_i)^2} \right)$$

N_A = doping concentration in p-region
 N_D = doping concentration in n-region

V_0 typ 0.6 – 0.9 V at room temp

Minority carrier drift

Once the electric field is established, some of the thermally-generated carriers drift towards the junction C get swept away, into the opposite side of the depletion region.



The total drift current, I_S , is the sum of:

- (1) the drift of thermally-generated holes in the n-region into the p-region
- (2) the drift of thermally-generated electrons in the p-region into the n-region

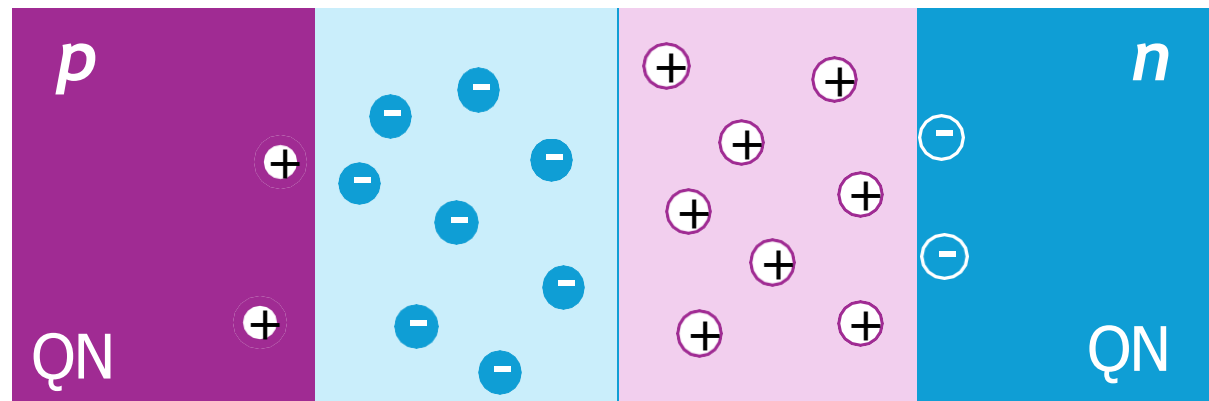
Under open-circuit conditions, the magnitudes of diffusion current = drift current.

$$I_D = I_S$$

Why does it stabilize?

Say $I_D > I_S$. More carriers diffuse to opposite sides

V_0 C Electric field increases, causing a slowdown in I_D



I_D = diffusion current

Eventually, $I_D = I_S$ (equilibrium)

Total currents in a semiconductor

Recall: In equilibrium (open circuit conditions) total current must be zero (KCL)

$$\text{For electrons: } J_{n,\text{drift}} + J_{n,\text{diffusion}} = 0$$

$$\text{or, } J_{\text{drift}} = -J_{\text{diffusion}}$$

- So any disturbance (e.g. light, doping gradient, thermal gradient) which may set up a carrier concentration gradient, will also internally set up a built-in current